Stable ∞-categories, Weinstein manifolds, and the Fukaya category

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April 8, 2022

Overview

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- The Fukaya category will be a functor

$$\left\{ \begin{array}{c} \text{some category of} \\ \text{symplectic manifolds} \end{array} \right\} \xrightarrow{\mathrm{Fuk}(-;\mathbb{Z})} \mathrm{Cat}^{\mathrm{st}}_{\infty,\mathbb{Z}}$$

Category theory review

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- An A_∞-category is a (1-)category enriched in chain complexes of graded R-modules where composition of morphisms is associative only up to higher homotopies
- Pretriangulated A_∞-categories model R-linear stable ∞-categories

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Morse's Lemma

Let $c \in Q$ be a critical point of a morse function f. Then there exists a coordinate chart $\phi : \mathbb{R}^n \hookrightarrow Q$ such that $\phi(0) = c$ and when restricted to this chart, f has the form

$$f \circ \phi(x) = f(c) - \sum_{i=1}^{k} x_i^2 + \sum_{j=k+1}^{n} x_j^2.$$

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- k is a number independent of the choice of chart called the *Morse index* of f at the critical point c

Morse theory and homotopy theory

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- Comes from the way the topology of the sublevel sets changes as they pass through a critical point
- Given a smooth function $f : Q \to \mathbb{R}$ and $t \in \mathbb{R}$, let $Q_t := f^{-1}(-\infty, t]$

Morse theory and homotopy theory

Proposition

Let $f : Q \to \mathbb{R}$ be a smooth function, and $a, b \in \mathbb{R}$ with $a < b$. If $f^{-1}[a, b]$ is compact and there are no critical values between a and b, then:

- the manifolds Q_a and Q_b are diffeomorphic
- Q_b deformation retracts onto Q_a

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Proposition

Let $f : Q \to \mathbb{R}$ be a smooth function and $c \in Q$ a non-degenerate critical point of f of index k. Write $v := f(c)$ and let $\epsilon > 0$ be a number such that $f^{-1}[v - \epsilon, v + \epsilon]$ is compact and c is the only critical point contained in it. Then the manifold $Q_{v+\epsilon}$ is homotopy equivalent to a space obtained from $Q_{v-\epsilon}$ by attaching a k-cell.

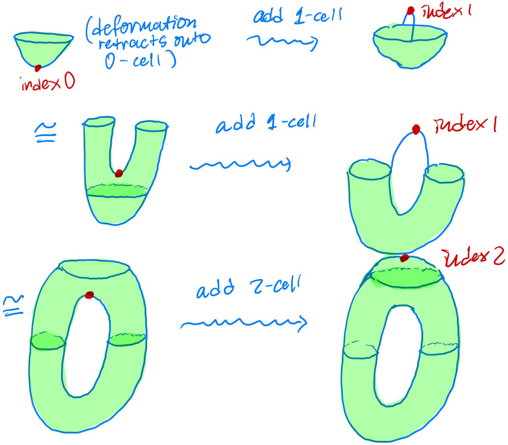
Morse theory and homotopy theory

Use the previous two propositions to show:

Proposition

Let $f : Q \to \mathbb{R}$ be a Morse function and assume that for each $t \in \mathbb{R}$, the manifold Q_t is compact. Then Q is homotopy equivalent to a CW complex with one k-cell for each critical point of f of index k.

Morse theory of the torus



Cell decomposition of the 2-dimensional torus via Morse theory

Symplectic manifolds

Question: What is a "CW complex" in symplectic geometry?

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Recall/Definition

A *symplectic manifold* is a pair (M, ω), where M is a $2n$-dimensional smooth manifold and $\omega \in \Omega^2_{\mathrm{dR}}(M; \mathbb{R})$ is a 2-form satisfying:

- The $2n$-form $\omega^{\wedge n}$ is a volume form on M (i.e. it is nowhere vanishing) or
- The map
$$TM \to T^*M$$
$$v \mapsto \omega(v, -)$$
is an isomorphism of vector bundles over M
- The 2-form ω is closed: $d\omega = 0$

Note: The first two conditions are equivalent

Symplectic manifolds

Observe that for (M, ω) a symplectic manifold, the second equivalent condition on the previous slide defines an isomorphism

$$\Gamma(M; TM) \xrightarrow{\cong} \Omega^1_{\mathrm{dR}}(M; \mathbb{R})$$

between vector fields on M and differential 1-forms

Example: the cotangent bundle

- Let Q be an n-manifold. Then T^*Q has a natural symplectic form $dp \wedge dq$ which is defined locally, so we can take $Q = \mathbb{R}^n$

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- Write coordinates of $\mathbb{R}^n$ as $(r_1, \ldots, r_n)$ and coordinates of $T^*\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$ as $(q_1, \ldots, q_n, p_1, \ldots, p_n)$

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- λ is often called the *tautological 1-form* or *Liouville 1-form*

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Definition

A symplectic manifold (M, ω) is *exact* if the symplectic form ω is exact, i.e. if there exists a 1-form λ such that $\omega = d\lambda$. In this case, the 1-form λ is called a *Liouville 1-form* or a *primitive for* ω. An *exact structure* on a symplectic manifold is a choice of primitive λ for ω.

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and the fact that $d\omega = 0$, we see that $d\lambda = \omega$ if and only if $\mathscr{L}_X \omega = \omega$

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Definition

A vector field X is called a *Liouville vector field* if $\mathscr{L}_X\omega = \omega$.

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- Given a Morse decomposition of a manifold Q, by applying T^* to the cells that make up Q, we can build up T^*Q symplectically

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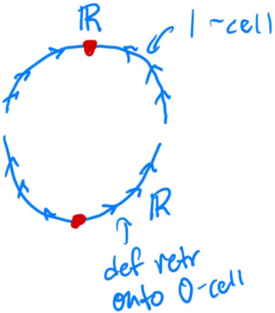
- More generally, we should be able to build symplectic manifolds out of "Weinstein cells"

Example: S^1



Vector fields on the cotangent bundle flow from lower index critical points to higher index critical points

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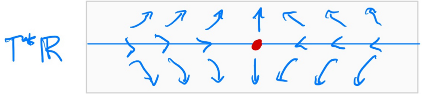
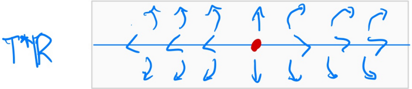
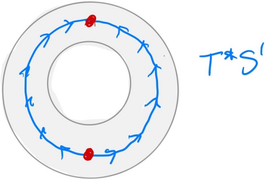




The cotangent bundles of the two cells agree at the boundary, so glue along the boundary



Weinstein manifolds

(Rough) Definition

A *Weinstein manifold* consists of

- a symplectic manifold (M, ω)
- A Liouville vector field for (M, ω) (or, equivalently, a primitive λ for ω)
- A Morse function $f : M \to \mathbb{R}$

along with some additional properties and compatibilities between the vector field X and the Morse function f.

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Answer: Weinstein manifolds play the role of CW complexes in symplectic geometry

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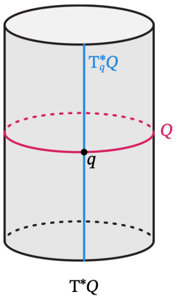
If M is a surface $(n = 1)$, then any curve is automatically Lagrangian. Since ω is a 2-form, it vanishes when restricted to any 1-dimensional submanifold.

Example: cotangent bundle



Example

Let Q be a manifold and write $\pi : T^*Q \to Q$ for the projection. The zero section $Q \subset T^*Q$ is Lagrangian, as is the cotangent fiber $T_q^*Q := \pi^{-1}(q)$ for any fixed $q \in Q$.

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- The compatibility requirement with the symplectic structure is that $\omega(-, J(-))$ defines a Riemannian metric on M
- Compatible almost complex structures on (M, ω) are unique (up to a contractible space of choices)

The Fukaya category

(Rough) Definition

Let (M, ω) be a symplectic manifold with a compatible almost complex structure $J : TM \to TM$. The *Fukaya category* $\mathrm{Fuk}(M; \mathbb{Z}/2)$ of M is a (pretriangulated) $\mathbb{Z}/2$-linear A_∞-category with

- objects Lagrangian submanifolds $L \subset M$ equipped with some extra structure
- given Lagrangians $L_0, L_1 \subset M$, deform L_1 so that the intersection is transverse. Then define the mapping complex as

$$\mathrm{Hom}(L_0, L_1) : \bigoplus_{x \in L_0 \pitchfork L_1} \mathbb{Z}/2[|x|],$$

  where $[|x|]$ denotes shifting by the degree of x.

The Fukaya Category

- The differential is given by

$$d(x) = \sum_y \# \left\{ \begin{array}{c} \text{pseudoholomorphic} \\ \text{disks from } x \text{ to } y \end{array} \right\} y.$$

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$$\overline{\partial} u = \frac{1}{2}(du + J \circ du \circ j) = 0,$$

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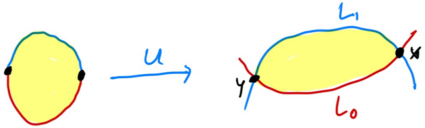

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is given on simple tensors by

$$m_k(p_k \otimes \cdots \otimes p_1) = \sum_{q \in L_0 \pitchfork L_k} \# \left\{ \begin{array}{c} \text{pseudoholomorphic polygons} \\ \text{with vertices } p_1, \dots, p_k, q \end{array} \right\} q$$

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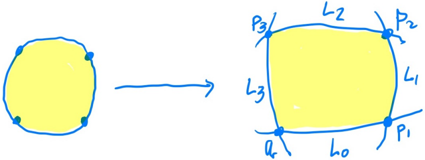

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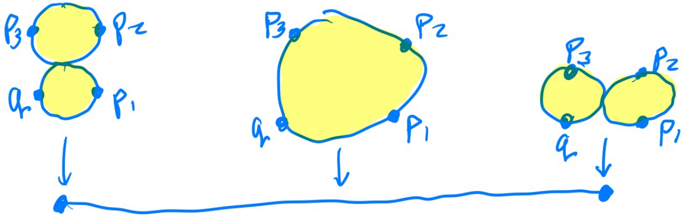
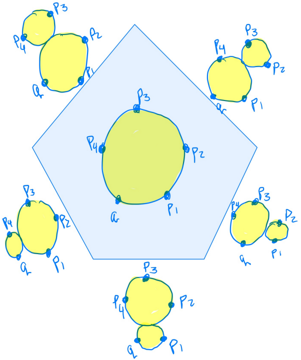
- highest dimensional face corresponds to the grouping into one disk, codimension one faces correspond to groupings with two disks, and so on
- the homotopies that contract these compactified moduli spaces are those that make the composition associative "up to higher homotopy"

Example: 4 points



So, the compactified moduli space is homeomorphic to a closed interval, which is a contractible 1-dimensional polyhedron

Example: 5 points



So, the compactified moduli space is homeomorphic to a pentagonal disk, which is a contractible 2-dimensional polyhedron

The category Wein

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Definition

The category of Weinstein manifolds Wein has as objects Weinstein manifolds and as morphisms codimension 0 embeddings $i : M \hookrightarrow N$ such that

$$i^* \lambda_N = \lambda_M + d \begin{pmatrix} \text{compactly supported} \\ \text{function} \end{pmatrix}.$$

The functor $\mathrm{Fuk}(-;\mathbb{Z})$

Theorem (Oh-Tanaka, Lazarev-Sylvan-Tanaka)

The ($\mathbb{Z}$-linear) Fukaya category defines a functor

$$\mathrm{Fuk}(-;\mathbb{Z}) : \mathrm{Wein} \to \mathrm{Cat}^{\mathrm{st}}_{\infty,\mathbb{Z}}$$

from Weinstein manifolds to $H\mathbb{Z}$-linear stable ∞-categories.

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- There is a modification $\mathrm{Wein}^{\diamond}$ of the category of Weinstein manifolds where we identify M with $M \times T^*\mathbb{R}^N$
- $\mathrm{Fuk}(-;\mathbb{Z})$ actually factors through this category

The mod p Moore space

Definition

The *mod p Moore space* $M(\mathbb{Z}/p, n)$ is a CW complex which has non-trivial reduced homology group $\mathbb{Z}/p$ only in degree n and is simply connected if $n \geq 2$. It is formed by attaching an $n+1$-cell along a degree p map $S^n \to S^n$ via the following pushout:

$$\begin{tikzcd} S^n \arrow[r, "p"] \arrow[d, hook] & S^n \arrow[d] \\ D^{n+1} \arrow[r] & M(\mathbb{Z}/p, n) \end{tikzcd}$$

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We can build a “regular Lagrangian disk” D_p out of a wedge of mod p Moore spaces

p-localization via symplectic geometry

Theorem (Lazarev-Sylvan-Tanaka)

The functor $- \times T^*D^n \backslash D_p$ is a symmetric monoidal localization of $\mathrm{Wein}^\diamond$, where D^n is any regular Lagrangian disk.

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- The endomorphism object of the unit $R := \mathrm{Map}_{H\mathbb{Z}}(\mathbb{1}, \mathbb{1})$ will be an E_∞-$H\mathbb{Z}$-algebra spectrum

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Theorem (Lazarev-Sylvan-Tanaka)

The functor $- \times T^*D^n \backslash D_p$ is a symmetric monoidal localization of $\mathrm{Wein}^\diamond$, where D^n is any regular Lagrangian disk.

- For any E_∞-algebra M in $\mathrm{Wein}^\diamond$, $\mathrm{Fuk}(-;\mathbb{Z})$ will send M to a symmetric monoidal stable ∞-category $\mathrm{Fuk}(M;\mathbb{Z})^\otimes$
- The endomorphism object of the unit $R := \mathrm{Map}_{H\mathbb{Z}}(\mathbb{1}, \mathbb{1})$ will be an E_∞-$H\mathbb{Z}$-algebra spectrum

Theorem (Lazarev-Sylvan-Tanaka)

The endomorphism object of the unit of the symmetric monoidal stable ∞-category $\mathrm{Fuk}(M \times T^*D^n \backslash D_p;\mathbb{Z})^\otimes$ is $R[\frac{1}{p}]$, the endomorphism object of the unit of $\mathrm{Fuk}(M;\mathbb{Z})^\otimes$ localized at p.